

INFO2602 Project Report

- **Group Name: Doh Spend**
- **Application Title: SmartSpender**

Members Table

Name	Student ID	GitHub Username	Role
Darryl Augustine	816039124	Augustine-dev	Marketing
Rondell Ramsundar	816044173	Imaoafkk	Lead, Devops, Uidesign

1. Project Summary

SmartSpender is a web-based dashboard application designed to help users monitor and manage their financial activity in a clear, holistic and structured way. The system is centralized, allowing users to input their monthly income, one-time expenses, and recurring subscription costs in order to better understand their financial habits and overall spending behavior.

A key feature of the application is the analytics section. By presenting this data through reports and analytics, users can make more informed financial decisions and avoid overspending.

The application also allows users to create their own custom budget further aiding in the organisation of every user's finances.

The application supports tracking and management of **one-time expenses**, **recurring subscriptions**, and **budget allocations**, all within a centralized interface. Additionally, users can view detailed reports and analytics that summarize their financial data, helping them identify trends, control expenses, and plan for future financial goals.

Overall, the system aims to promote financial awareness, discipline, and smarter money management through simple yet effective tools and visual insights.

2. Features Implemented

Feature 1: Expense Tracking

Description:

Input monthly salary as well as expenses earned and paid to be analyzed.

Feature 2: Subscription Tracking

Description:

Track recurring subscriptions (Netflix, Spotify, etc.) with next billing dates.

Feature 3: Budget Creation

Description:

Input a category of expense and the amount to monitor spending and flag over spending

Feature 4 : Report Creation

Description:

Use inputted expenses to display analytics

3. UI Design

3.1 Wireframe Diagrams

- Screen 1: Dashboard

S\$

Dashboard

April 2026

Monthly Salary

+Add

\$5,200.00

Update Salary

Total Income

\$6,100

Total Expenses

\$3,420

Net savings

\$2,680

Recent Transactions

+Add

Apr 12Netflix-\$17

Apr 11Salary+\$900

Apr 10Groceries-\$84

Subscriptions

+Add

NFNetflix\$17/mo

SPSpotify\$10/mo

- Screen 2: Transactions(with filter bar)

S\$

Search transactions...

All types

All cats

+Add

Date	Name	Category	Type	Amount
Apr 12	Netflix	entertainment	expense	Delete
Apr 11	Freelance	other	income	Delete
Apr 10	Groceries	food	expense	Delete
Apr 9	Uber	transport	expense	Delete
Apr 8	Electric Bill	bills	expense	Delete

- Screen 3: Budget

S\$

☐
☐
☒
☐

Monthly Budget Limits

+Set Budget

Food

\$148 / \$300

Transport

\$78 / \$100

Entertainment

\$130 / \$120

Bills

\$210 / \$400

Health

\$0 / \$50

- Screen 4: Analytics

S\$

☐
☐
☐
☒

Total Expenses

\$3,420

Bills 35%

Food 22%

Entertain 18%

Spending category

Bills

Food

Entertainment

Transport

Shopping

\$1,200

\$752

\$614

\$378

\$256

Monthly Savings

44% savings rate

Savings

\$2,680

44% of total Income

Income vs Expenses

Expense Breakdown

- Screen 5: Add Transaction Modal

s\$

Add Transaction

Name

Amount

Type Category

Date

Cancel Save

3.2 Typography

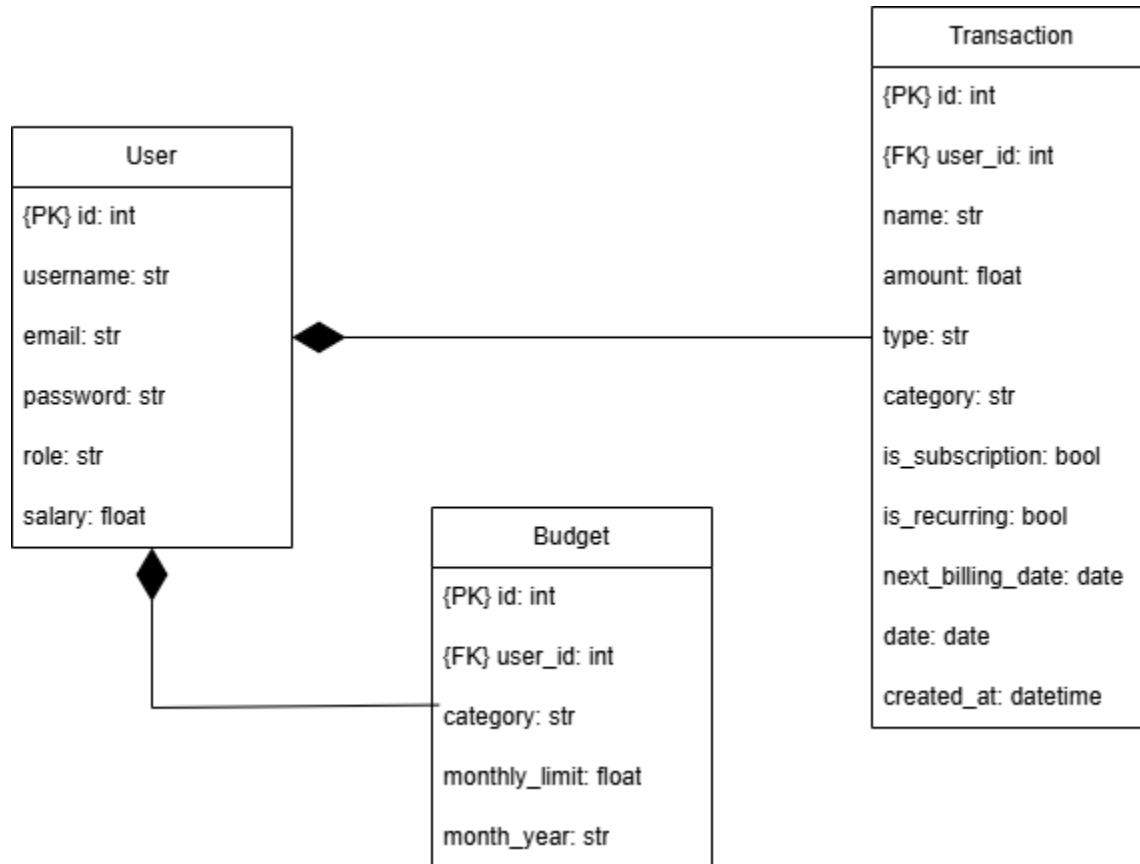
- Primary Font: Inter
- Secondary Font: Siegf UI / Roboto / sans-serif
- Usage (headings, body text, buttons, etc.): Inter is used for all text elements including headings, body text, buttons, labels, and numbers. Segoe UI, Roboto, and generic sans-serif act as fallback fonts if Inter is unavailable

3.3 Colour Palette

Purpose	Colour	Hex Code
Primary	Teal Green	#00D4AA
Secondary	Purple	#6C5CE7
Accent	Blue	#0984E3
Background	Deep Navy	#0A0A1A
Text	White	#FFFFFF

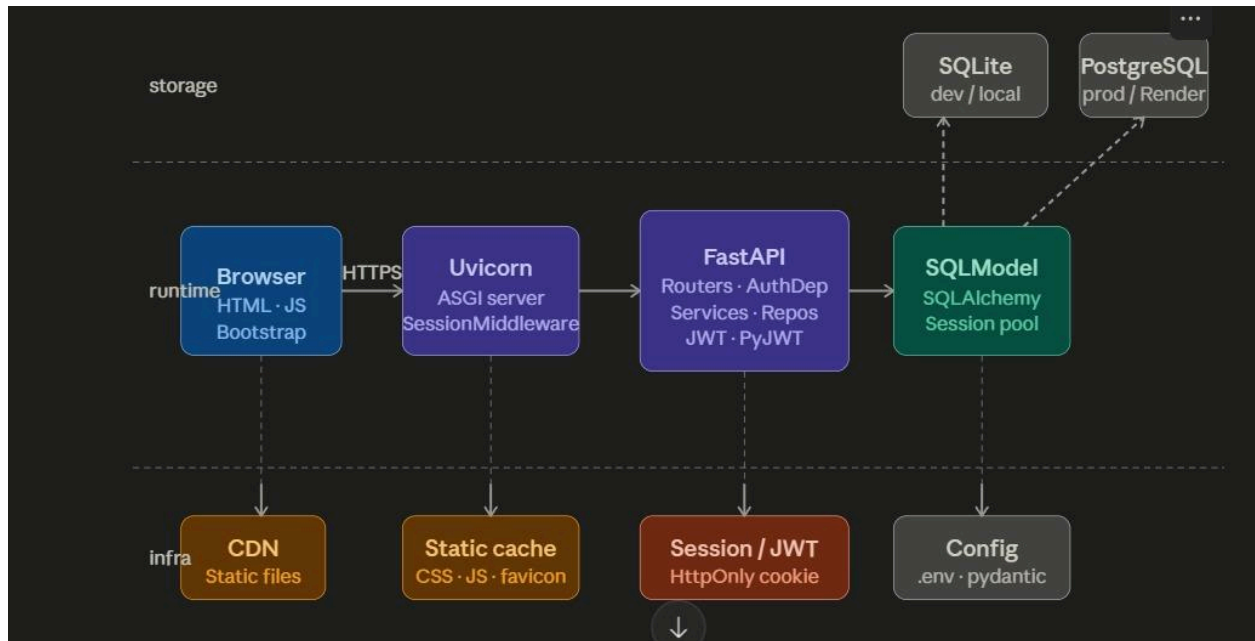
4. Model Design

4.1 Class Diagram (SQLModel)



5. Architecture

5.1 System Design



- Frontend: User interface built with Jinja2, Bootstrap, JavaScript, and Highchart
 - Backend: FastAPI application that handles routing, authentication, and finance
 - Database: SQL database storing users, transactions, and budgets
 - Data Flow: Browser sends requests to backend, backend processes data and reads and writes to the database, then returns results to the user
-

6. APIs, Technologies, Resources

6.1 APIs Used

API Name	Purpose
HighCharts	uses Highcharts for financial data visualization
Google Fonts API	Loads the Inter font and Symbols icons

6.2 Technologies

Technology	Purpose
FastAPI	Backend web framework used to build the application, handle routing and process requests
SQLModel	ORM used to define models and interact with SQL database
SQLite	Relational database to store users, transactions and budgets
Jinja2	Template engine used to render dynamic HTML pages
JavaScript	Uses client side interactivity, API calls, and updating the interface dynamically
Bootstrap	Uses the responsive layout to build forms, buttons, tables, and modal components
Render	Cloud hosting platform used to deploy the application

CSS3	Styles the application and implement the dark dashboard theme
HighCharts	Generates analytics and financial data visualizations
HTML5	Structures the frontend pages
PostgreSQL	Uses the production SQL database on Render

6.3 Resources / Templates

Resource	Link / Description
HighCharts	https://api.highcharts.com/highcharts/
GitHub	https://github.com/
Render	https://render.com/
Google Fonts	https://fonts.google.com
Bootstrap	https://getbootstrap.com/
Project Starter Template	https://github.com/uwidi/FastAPIStarter

7. Screenshots

SmartSpender
Finance Tracker

Dashboard

Transactions

Recurring

Budget

Analytics

Refresh data

bob

Logout

Monthly Salary
\$2000.00
+ Update Salary

Total Income
\$2,100.00

Total Expenses
\$859.00

Net Savings
\$1,241.00

Recent Transactions

DATE	NAME	CATEGORY	AMOUNT	
2026-04-15	Spotify	entertainment	-\$140.00	Delete
2026-04-14	Netflix	entertainment	-\$49.00	Delete
2026-04-14	Odd Job	hills	+\$100.00	Delete

Active Subscriptions

NAME	AMOUNT	NEXT BILLING	
Spotify	-\$140.00	2026-04-14	Cancel
Netflix	-\$49.00	2026-05-12	Cancel

SmartSpender
Finance Tracker

Dashboard

Transactions

Recurring

Budget

Analytics

Refresh data

bob

Logout

Update Monthly Salary

Monthly Salary (\$)

2000.0

Cancel Save

Total Income
\$2,100.00

Total Expenses
\$859.00

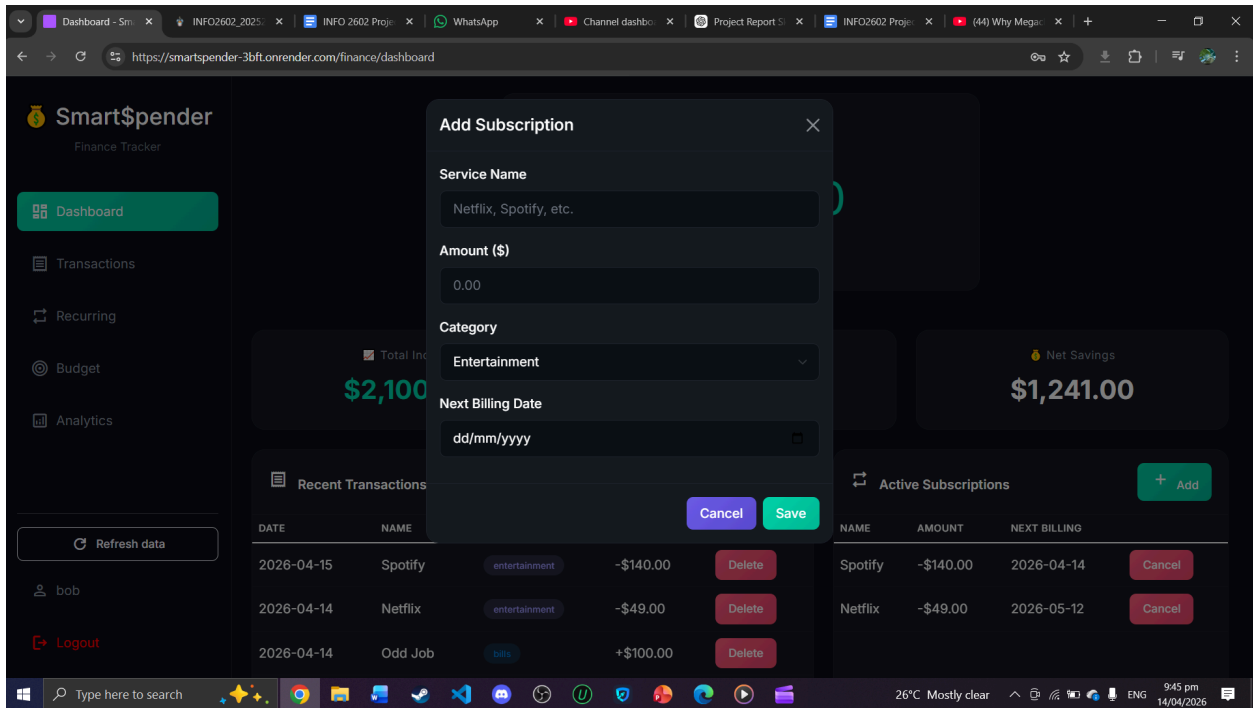
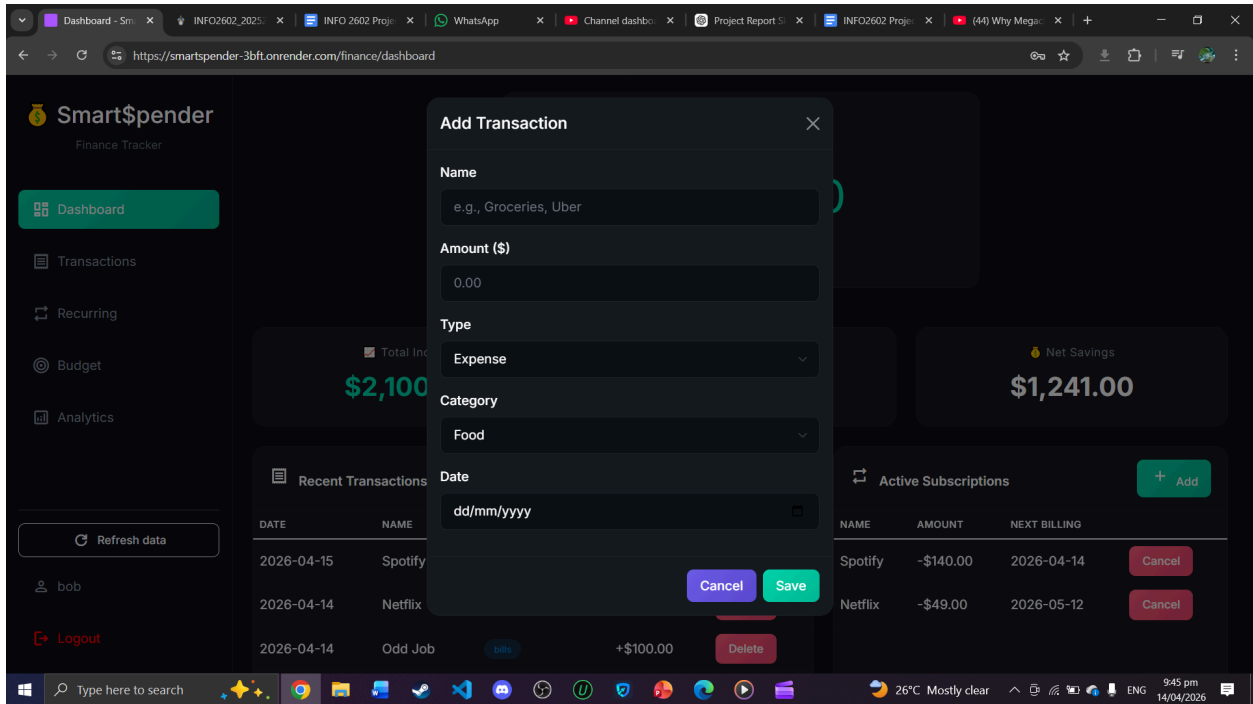
Net Savings
\$1,241.00

Recent Transactions

DATE	NAME	CATEGORY	AMOUNT	
2026-04-15	Spotify	entertainment	-\$140.00	Delete
2026-04-14	Netflix	entertainment	-\$49.00	Delete
2026-04-14	Odd Job	hills	+\$100.00	Delete

Active Subscriptions

NAME	AMOUNT	NEXT BILLING	
Spotify	-\$140.00	2026-04-14	Cancel
Netflix	-\$49.00	2026-05-12	Cancel



Smart\$ponder

Finance Tracker

Dashboard

Transactions

Recurring

Budget

Analytics

Refresh data

bob

Logout

All Transactions

Add Transaction

DATE	NAME	CATEGORY	TYPE	AMOUNT	ACTIONS
2026-04-15	Spotify	entertainment	expense	-\$140.00	Delete
2026-04-14	Netflix	entertainment	expense	-\$49.00	Delete
2026-04-14	Odd Job	bills	income	+\$100.00	Delete
2026-04-13	Groceries	food	expense	-\$400.00	Delete
2026-04-10	Elctric Bill	bills	expense	-\$200.00	Delete
2025-12-12	Uber	transport	expense	-\$70.00	Delete

26°C

9:45 pm

14/04/2026

Smart\$ponder

Finance Tracker

Dashboard

Transactions

Recurring

Budget

Analytics

Refresh data

bob

Logout

Subscriptions & Recurring Payments

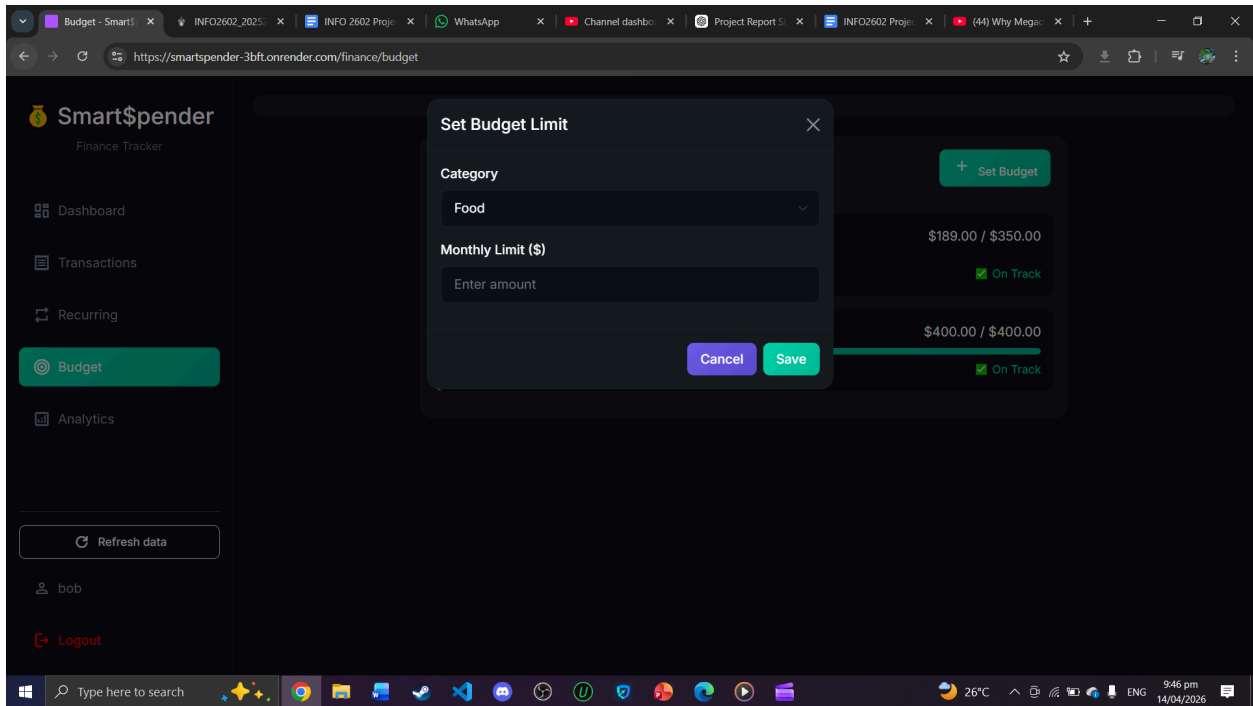
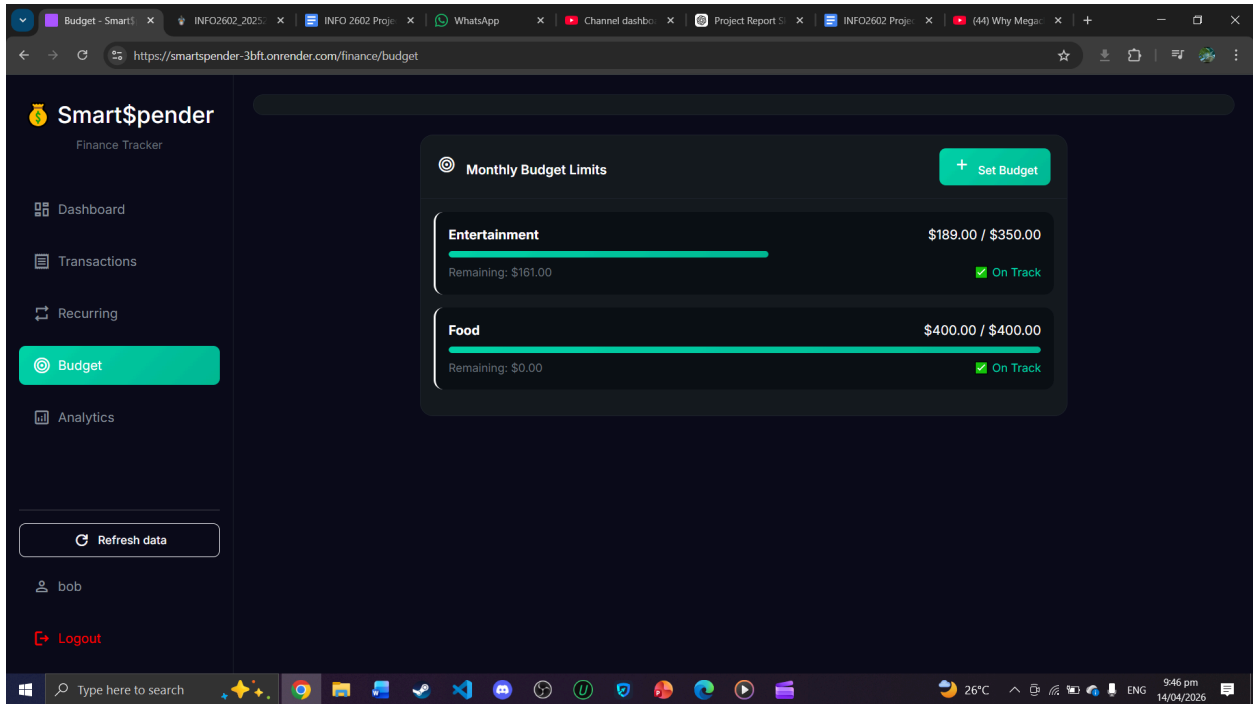
Add Subscription

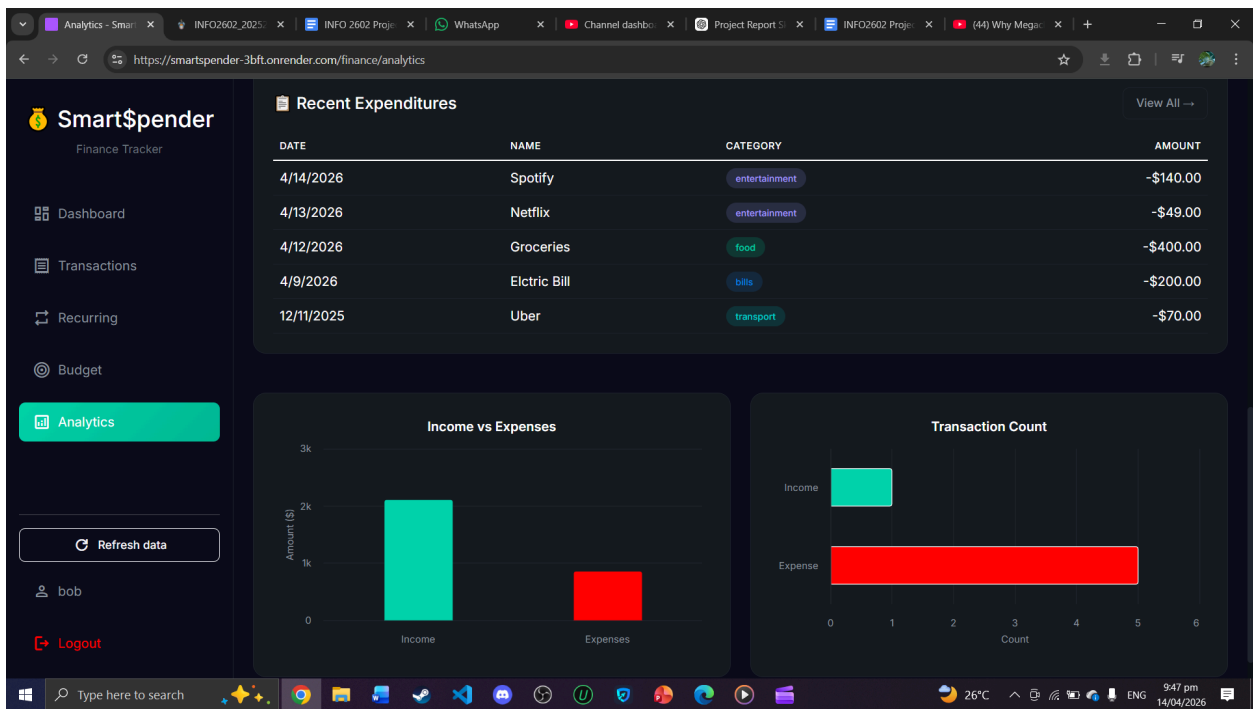
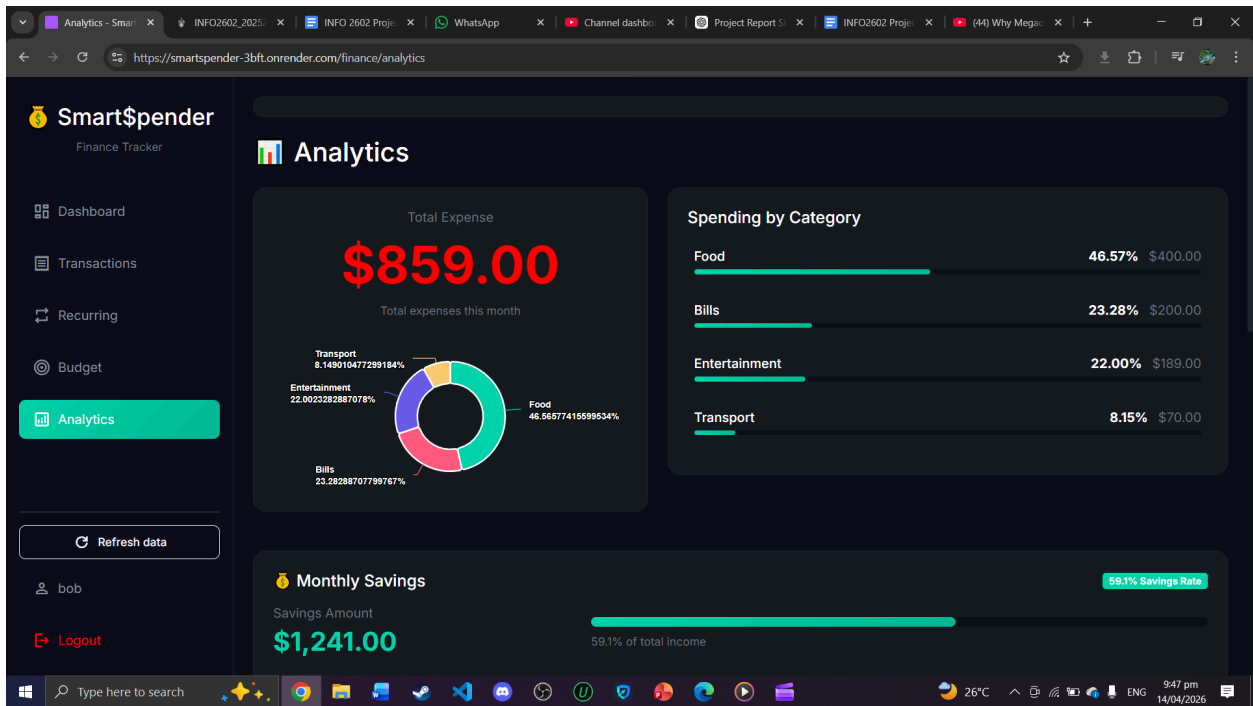
NAME	AMOUNT	CATEGORY	NEXT BILLING	ACTIONS
Spotify	-\$140.00	entertainment	2026-04-14	Cancel
Netflix	-\$49.00	entertainment	2026-05-12	Cancel

26°C Mostly clear

9:46 pm

14/04/2026





8. Deliverables

GitHub Repository Link :<https://github.com/lmaofkk/SmartSpender>

Presentation: <https://youtu.be/iLQF2RXh9n8>

Deploy Link: <https://smartspender-3bft.onrender.com/>

Login Credentials : Username - bob
Password - bobpass